MTH203: Sequences and series

Tutorial 01

12/08/2025

Problem 1. Construct a bijection $f : \mathbb{N} \to \mathbb{Z}$.

Problem 2. Let $y_1 = 6$ and for each $n \in \mathbb{N}$, define $y_{n+1} = (2y_n - 6)/3$.

(a) Use induction to show that this sequence satisfies $y_n > -6$ for all natural numbers n.

(b) Use induction to show that $y_n > y_{n+1}$ for all natural numbers n.

Problem 3. Let F be an ordered field. We denote order relation on F by $\geqslant$. We will write $x > y$ if $x \geqslant y$ and $x \neq y$.

(a) Prove that if $x \geqslant y$ and $z \geqslant w$, then $x + z \geqslant y + w$. Prove that in this case, we have $x + z = y + w$ if and only if $x = y$ and $z = w$.

(b) Prove that $x > 0$ if and only if $-x < 0$.

(c) Prove that if $x > 0$ and $y < 0$ then $xy < 0$.

(d) Prove that if $x < 0$ and $y < 0$ then $xy > 0$.

(e) Prove that $x^2 > 0$ for any $x \neq 0$.

(f) Prove that $-1 < 0$.

Problem 4. Let A be any set and suppose that $f : A \to \mathbb{N}$ is a surjective function. Prove that there exists a function $g : \mathbb{N} \to A$ such that $f(g(x)) = x$ for all $x \in \mathbb{N}$. What happens if we replace $\mathbb{N}$ by an arbitrary set B in this question?